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Subject–action–object-based morphology analysis for determining the direction of technological change

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ABSTRACT

Morphology analysis, despite being a strong stimulus for the development of new alternatives, largely relies on domain experts and neglects the relationships between keywords in the construction of morphological structures. In addition, there are few systematic approaches to prioritize the morphological configurations. To address these issues, a hybrid approach is proposed, which enhances the performance of morphology analysis by combining it with subject–action–object (SAO) semantic analysis. Initially, a keyword co-occurrence patent set for subsequent SAO analysis is prepared based on keywords frequency vector analysis. Then, SAO structures are extracted and semantic analysis is performed to identify the relationships between keywords, which help to build morphological structures more objectively. In addition, a well-defined evaluation system that contains eight sub-indexes is proposed to evaluate the morphological configurations. Finally, to demonstrate and validate the proposed approach, the dye-sensitized solar cells technology is employed as the case study. Results indicate that the most promising combination we predict appears frequently in 2012–2014 and the distribution of it is also close to the fact in 2012–2014. Accordingly, the proposed method can be used to effectively determine the direction of technological change and to forecast technology innovation opportunities.

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1. Introduction

Technology forecasting anticipates the direction and rate of technology change and thus facilitates decision making in managerial issues such as setting priority, resource allocation and risk reduction for technology development. Therefore, technology forecasting can satisfy both public and private needs (Yoon and Park 2007). Numerous techniques have been developed to support technology forecasting. Broadly, technology forecasting techniques can be divided into two categories: exploratory and normative methods. Morphology analysis, a normative technology forecasting method, is considered a strong stimulus for the development of new alternatives (Xu and Leng 2012; Yoon et al. 2008).

When compared with other methods, the strength of morphology analysis lies in its speed and ease of application because the process can be implemented and quickly evaluated and does not require excessive amounts of data (Yoon and Park 2007). It has two key elements: construction of morphological structure and evaluation of morphology configurations. However, morphology analysis cannot operate independently, i.e. it must either be supported by or used as a facilitator in support of the forecasting process of other technology forecasting methods. Therefore, in recent research, computerized algorithms and text mining technology have been introduced into morphology analysis to identify future technology opportunities (Yoon et al. 2014).

However, two aspects of current research require improvement. First, instead of a keyword-based method, a semantic-based method is required to build technology morphological structure. In previous research, the construction of morphological structure has been primarily dependent on expert knowledge and intuition. To reduce this dependency on domain experts and manual expertise, several studies have introduced keyword-based methods to construct morphological structure (Xu and Leng 2012; Yoon and Park 2004). However, a keyword-based method can only identify connections between keywords. It cannot clearly represent what kind of relationships they are and is unable to identify the purpose of a technology. Therefore, a method with semantic analysis is needed to enhance the accuracy of morphological structure.

Second, a systematic and comprehensive technology evaluation index system is needed to evaluate morphological configurations. Yoon and Park used patent citation analysis and conjunction analysis to prepare priority lists for morphological configurations (Yoon and Park 2007). Based on Yoon's method, Xu and Leng added the frequency of keywords and factor contribution to evaluate morphological configurations (Xu and Leng 2012). In fact, due to an absence of citations or citation delay, some patent technology might impose limitations during patent citation analysis. Therefore, based on citation analysis, a systematic and comprehensive technology evaluation index system is required to identify all feasible forms of new technology development and forecast the most promising combinations.

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To address these issues, this study proposes an improved morphology analysis approach for technology forecasting based on subject–action–object (SAO) semantic analysis and text mining method.

SAO structure is a key concept that can show the relationship between components used in a relevant patent (Cascini et al. 2004a). If we connected SAO structures in a chain fashion, the relationship between each S/O can be identified by analysing the action (A). Therefore, in the proposed method, we use SAO chain semantic analysis and text mining to construct technology morphological structure. Based on patent analysis and bibliometrics analysis, a well-defined process that utilizes quantitative data is proposed to support evaluation of morphological configurations. In this process, we synthetically build two indexes, a technology importance index and a technology development potential index, which include eight evaluation sub-indexes in total. Relative to data selection, patent documents and literature data are integrated for consideration. Note that the dataset is divided into three parts, i.e. training data, validating data and test data, to validate the proposed method.

This remainder of this paper is organised as follows. In Section 2, works related to morphology analysis and SAO semantics are defined. In Section 3, the concept and framework of the hybrid method is proposed and explained in detail. In Section 4, a dye-sensitized solar cell (DSSC) case is examined to illustrate and validate the proposed method. Finally, concluding remarks, limitations of the current research and future considerations are discussed in Section 5.

2. Background

2.1. Morphology analysis

Morphology analysis was introduced by Zwicky in the 1940s to structure and investigate the total set of relationships contained in multi-dimensional, non-quantifiable problem complexes (Zwicky 1969). It is a method to structure rather than solve a problem. A system is typically composed of a number of subsystems, each of which may be shaped in a number of different ways. Therefore, morphology analysis can be used to identify the various shapes that each dimension takes. By combining these shapes, it examines all possible alternatives that a system may adopt (Wissema 1976).

In traditional morphology analysis, the construction of morphological structure primarily depends on expert knowledge and intuition (Ritchey 1998). To reduce the dependency on domain experts and manual expertise, several computer-aided methods have been introduced to improve morphology analysis. Ritchey introduce advanced computer support for morphology analysis (Ritchey 2003). Meanwhile, other methods, such as scenario laboratories (Ritchey 2009), bayesian networks (Waal and Ritchey. 2007), are introduced into computer-aided morphology analysis for strategic decision support. Besides, Yoon and Park proposed a patent text keyword-based morphology analysis approach. They used conjoint analysis to eliminate infeasible combinations and list the remaining combinations in order of importance (Yoon and Park 2007). Yoon, Park and Coh further introduced text mining method into morphology analysis to explore technology opportunity (Yoon et al. 2014).

In these approach, the computerized algorithms and text mining technology help a lot for technology forecasting. However, the keyword-based method cannot represent how keywords are related and is unable to identify the purpose of a technology. Therefore, in this paper, we introduced SAO semantic analysis to compensate for the limitations of keywords-based approaches.

2.2. SAO semantic analysis

SAO structures are composed of Subject (noun phrase), Action (verb phrase) and Object (noun phrase). It emphasizes the "key concepts" and can provide various technology information on their semantic relationships (Cascini et al. 2004b; Choi et al. 2012a).

Generally, the SAO structures are extracted from technology literature or patent documents. Consequently, SAO structures can offer a significant amount of technology information. In particular, in SAO structures, subjects and objects may refer to components of a system, and actions may refer to functions performed by and on components (Choi et al. 2012b). Thus, this syntactically ordered sentence can explicitly describe a relationship between components that appear in the relevant article (Yoon and Kim 2011a). It can also state partitive relationships among products or technologies. Besides, an SAO structure can be organized in a problem–solution format if the action–object (AO) forms the problem and the subject (S) states the solution.

Several researchers have attempted to use SAO structures for meeting various needs, including inventor analysis (Moehrle et al. 2005), patent infringement risk analysis (Bergmann et al. 2008), patent similarity measure (Cascini and Zini 2008), technology monitoring (Chang et al. 2010), product forecasting (Gerken et al. 2010), construction of technical trees (Choi et al. 2012c), technology trends analysis (Yoon and Kim 2011a; Choi et al. 2011), patent function network analysis (Choi et al. 2010), R&D planning (Yoon and Kim 2011b) and technology opportunity analysis (Yoon and Park 2005; Yoon and Kim 2012).

With its unique advantages, SAO semantic analysis can provide more valuable information than a keyword-based method. Therefore, in this study, the SAO semantic analysis method is used for the construction of morphological structure.

3. Methodology

3.1. Basic concepts

In this paper, to address the limitations of traditional keyword-based method, we introduce SAO method to determine the relationship between product and components/materials. For example, a typical case of SAO structure is 'thin film cells composed of MnPc and monoazo dye'. In the sentence, 'thin film cells' is the product, and 'MnPc' and 'and monoazo dye' are its materials. Thus, the relation between product and material can be identified.

The SAO-based morphology analysis approach for technology forecasting primarily includes four modules (Fig. 1). The basic concept is as follows.

First, based on keywords frequency vectors, we identify a patents dataset that includes two or more keywords in a patent text. We then extract SAO structures from these patent texts and connect SAO structures based on the similar words from the subject and object. Thus, we can identify the relationship of keywords based on the action (A). Simultaneously, the SAO structures are extracted from the entire text of selected patents. Therefore, the words beyond the list of keywords may also appear in SAO structures. Thus, the words and relationship are not confined to the list of keywords. Note that some low and

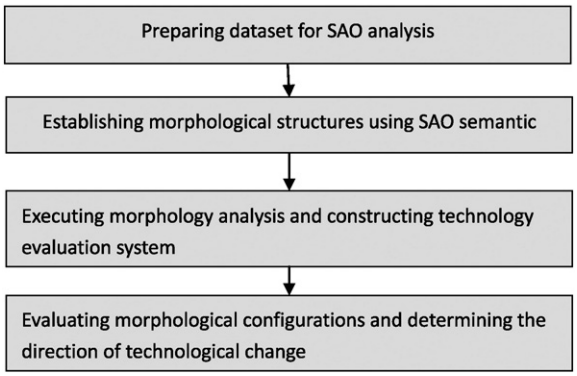


Fig. 1. Framework of the SAO-based morphology analysis approach.

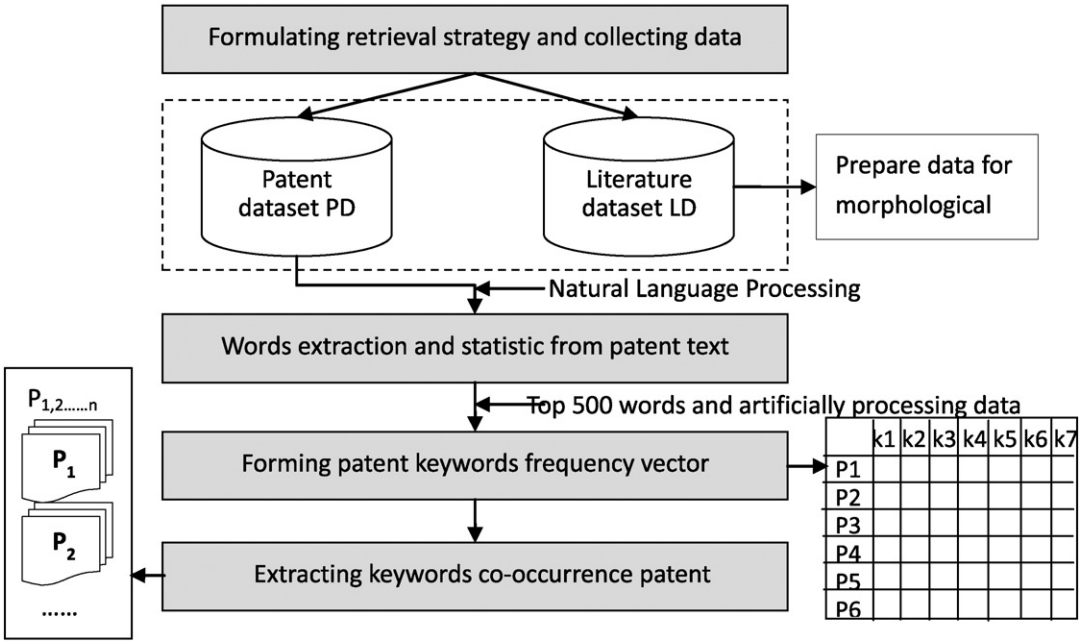


Fig. 2. Preparing dataset for SAO analysis.

intermediate frequency important words and their relationships can also be identified. This is very important for the construction of technology morphological structure.

Second, to forecast future technology morphological configurations, we connect literature data and patent data to construct a morphological evaluation system, combine bibliometrics methods with regression analysis to establish a morphological configuration evaluation model and use historical data to verify the accuracy of the prediction model.

Consequently, this systematic approach can compensate for the previously discussed drawbacks of morphology analysis. This results in better forecasting of future technology morphological configurations and identification of technology innovation opportunities.

3.2. Framework for SAO-based morphology analysis

3.2.1. Preparing dataset for SAO analysis

This module primarily collects data and screens a patent set for subsequent SAO analysis. This procedure has four steps, as shown in Fig. 2.

First, we select a technology field and formulate a retrieval strategy to collect data. Second, using natural language processing, we statistically determine word and phrase (hereafter referred to as words) frequency. We then select the top 500 high-frequency words and manually remove the irrelevant words. Thus, the retrained words are considered keywords. Third, patent abstracts are transformed into keywords frequency vectors using natural language processing technology, which can be used for patent screening and to filter keywords further. Fourth, based on keywords frequency vector analysis, we screen patents that include two or more keywords in abstract. The retrained patent are considered keywords co-occurrence patent set.

The purpose of this study is to forecast future potential technology morphological configurations and to identify technology opportunity based on morphology analysis methods. In morphology analysis, the morphological configurations are identified by analyzing keyword frequency vectors. Therefore, to obtain satisfactory results, complete patent documents are required to perform morphology analysis. Therefore, we have selected the U.S. Patent and Trade Office (USPTO) database for patent documents. In addition, the Science Citation Index Expanded (SCI-Expanded) database is required to evaluate morphological configurations, which can supplement the patent evaluation indexes.

3.2.2. Establishing morphological structure using SAO semantic analysis

The morphological structure is established by SAO structure analysis. Note that the morphological structure for morphology analysis is a hierarchy technology tree. Therefore, we collect action words that mean containing (hereafter referred to as partative words). Then, we extract SAO structures from the selected patent text using the Stanford Parser software and screen SAO structures according to the partitive words. We then connect SAO structures based on the similar word from the S/O. Thus, several SAO structures can form a SAO chain. Then, the hierarchy relationship among different S/O can be identified by analysing the type of A (Table 1).

Table 1
Hierarchy of SAO chains.

Number	Type of SAO chain	Action type	Level 1	Level 2	Level 3
1.1	$S–A_1–O_1$	A1 and A2 both mean containing	S	O_1,O_2	
1.2	$(S)–A_2–O_2$	A1 and A2 both mean contained	O_1,O_2	S	
1.3		A1 means containing; A2 means contained	O_2	S	O_1
1.4		A1 means contained; A2 means containing	O_1	S	O_2
2.1	$S_1–A_1–(O)$	A1 and A2 both mean containing	S_1,S_2	O	
2.2	$S_2–A_2–(O)$	A1 and A2 both mean contained	O	S_1,S_2	
2.3		A1 means containing; A2 means contained	S_1	O	S_2
2.4		A1 means contained; A2 means containing	S_2	O	S_1
3.1	$S_1–A_1–(O)$	A1 and A2 both mean containing	S_1	O	O_2
3.2	$(S)–A_2–O_2$	A1 and A2 both mean contained	O_2	O	S_1
3.3		A1 means containing; A2 means contained	S_1,O_2	O	
3.4		A1 means contained; A2 means containing	O	S_1,O_2	

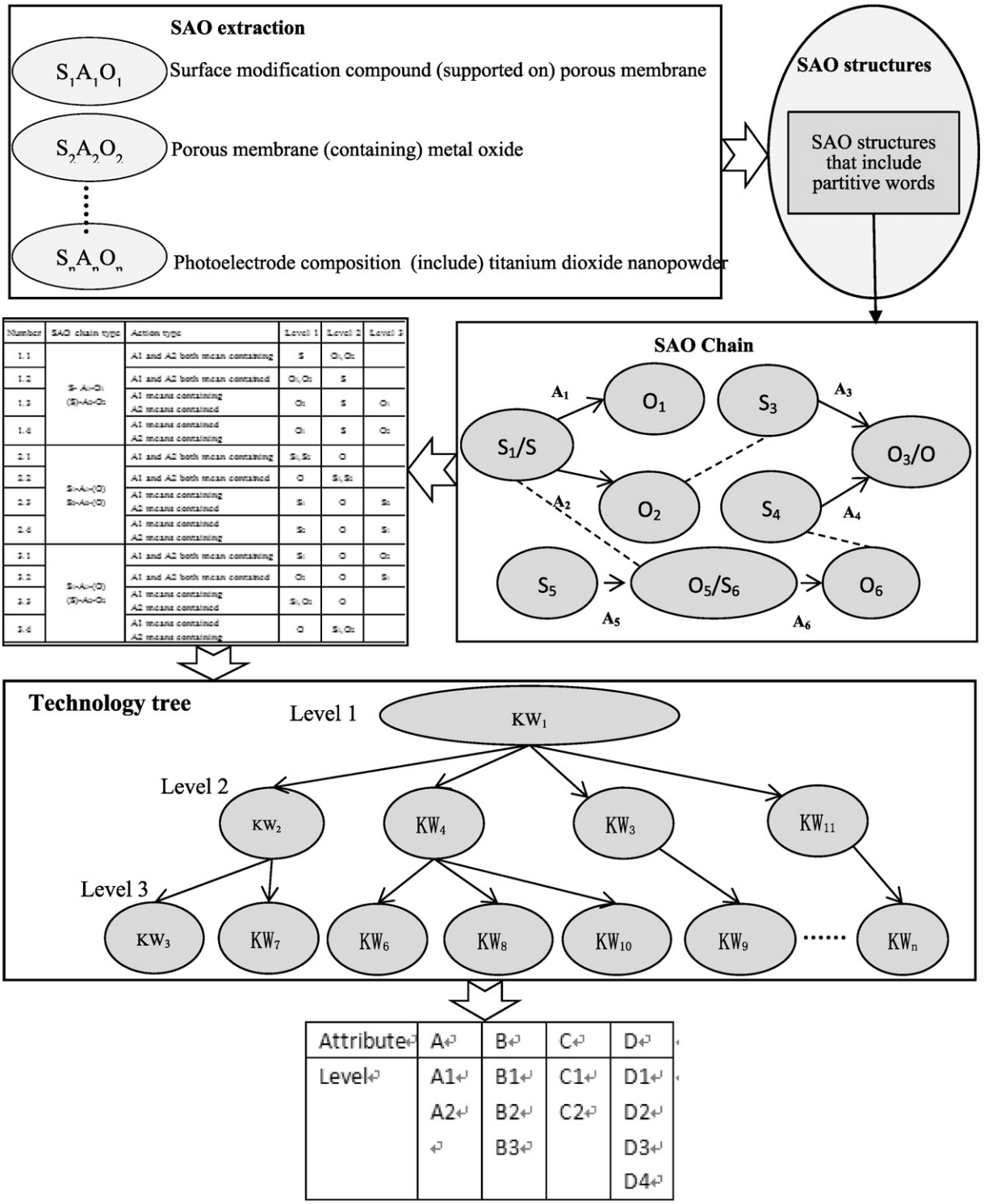
Finally, the technology hierarchy tree can be created based on these keyword relationships. Thus, by artificially merging and filtering, we can establish a technology morphological structure. This procedure is shown in Fig. 3.

3.2.3. Executing morphology analysis and constructing evaluation system

Based on the morphological structure discussed in Section 3.2.2, each patent document is transformed into a vector according to the list of keywords. Then, the morphology of each patent is identified by calculating and matching keywords in the morphological structure. This calculation process is shown in Fig. 4.

First, we first calculate the value of levels in each attribute based on keywords vector, which is the sum of the corresponding keywords frequency. Second, we use the max level to present the corresponding attribute. In this way, the morphology combination of a patent can be obtained. In Fig. 4, the final morphology combination is A2–B1–…–Nn.

Unoccupied technology configurations can be derived by observing the morphology of existing patents. To identify the technological



Attribute	A	B	C	D
Level	A1	B1	C1	D1
	A2	B2	C2	D2
		B3		D3
				D4

Fig. 3. Establishing morphological structure based on SAO semantic analysis.

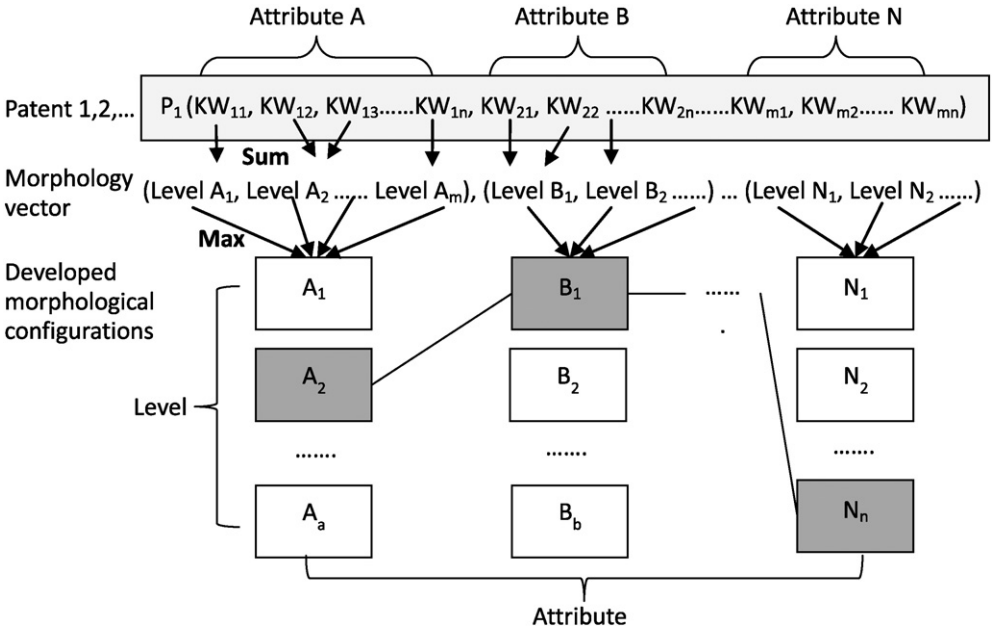


Fig. 4. Calculating and matching keywords into morphological structure.

feasibility of unoccupied technology configurations and forecast promising future technological configurations, we construct a comprehensive index system to evaluate the utility of each level in the morphological structure. Considering that literature data contains information about future technology development, we combine a literature dataset with a patent dataset to construct the technology evaluation system, which includes eight evaluation sub-indexes (six indexes to evaluate technology importance and two indexes to evaluate future technology potential). These evaluation indexes are detailed in Table 2.

3.2.4. Evaluating morphological configurations and determining the direction of technological change

In this module, one of the main goals is to build a technology morphology forecasting model. To validate the proposed method, we divide the dataset into three parts, subset 1 (1991–2009), subset 2 (2009–2011) and dataset 2 (2012–2014). Note that subset 1 and subset 2 are combined to form dataset 1. Dataset 1 is used to execute the entire process of the proposed method, and Dataset 2 is used to verify the validity of the proposed method.

The process of building the technology morphology forecasting model is shown in Fig. 5. First, we identify the unoccupied morphological configurations in subset 1 (1991–2009) and new morphological configurations in dataset 2 (2012–2014). Second, the patent data and literature data in subset 1 (1991–2009) is used to calculate the value of each index, which is the independent variable of forecasting model. Subset 2 (2009–2011) is used to calculate the frequency of each level in the existing morphological configurations, which is as the dependent variable. The formula for the technology morphology forecasting model can then be obtained using regression analysis. Finally, based on the morphology forecasting model, we use patent dataset 1 to obtain the utility score for each level. Thus, the utility score of the morphological configurations can be calculated and the most promising future morphological configurations can be identified.

4. Technology opportunity analysis of DSSCs

4.1. Data preparation

To illustrate the feasibility of the proposed method, we conducted an empirical study using patent and literature data for DSSC technology.

First, 1879 patent documents related to DSSC technology were retrieved from the USPTO database, and 7884 literature documents (document type = article) from the SCI-Expanded database were collected. The retrieval strategy used was developed by Ying, Ting and Alan (Guo et al. 2012).

Using the software tool VantagePoint from Search Technology Inc. to pretreat this data, we obtained a word-frequency list. Then, from 500 high-frequency words, we manually removed the irrelevant words, to obtain 166 technology keywords (Table 3). We then transformed patent abstracts into keywords frequency vectors and formed an 1879 × 166

Table 2
Technology evaluation index system to evaluate the utility of levels.

Type	No	Evaluating index	Explanation
Technology importance index(TI)	TI_1	The year technology appear	Year(latest)-Year(appear first)
	TI_2	Average time for patent authorization	Sum(pub.year-appl.year)/N
	TI_3	Keywords frequency of patent	The keywords occurrence frequency of each level(Patent)
	TI_4	Keywords frequency of literature	The keywords occurrence frequency of each level(Literature)
	TI_5	Total cited frequency	Sum(citation frequency of patent containing the level)
	TI_6	Average citation frequency	Sum(citation frequency of patent containing the level)/N
Technology potential index(TP)	TP_1	Recent 5 year keywords frequency	The keywords occurrence frequency of each level in recent 5 year (Literature)
	TP_2	Recent 5 year citation frequency	Sum(citation frequency of patent containing the level in recent 5 year)

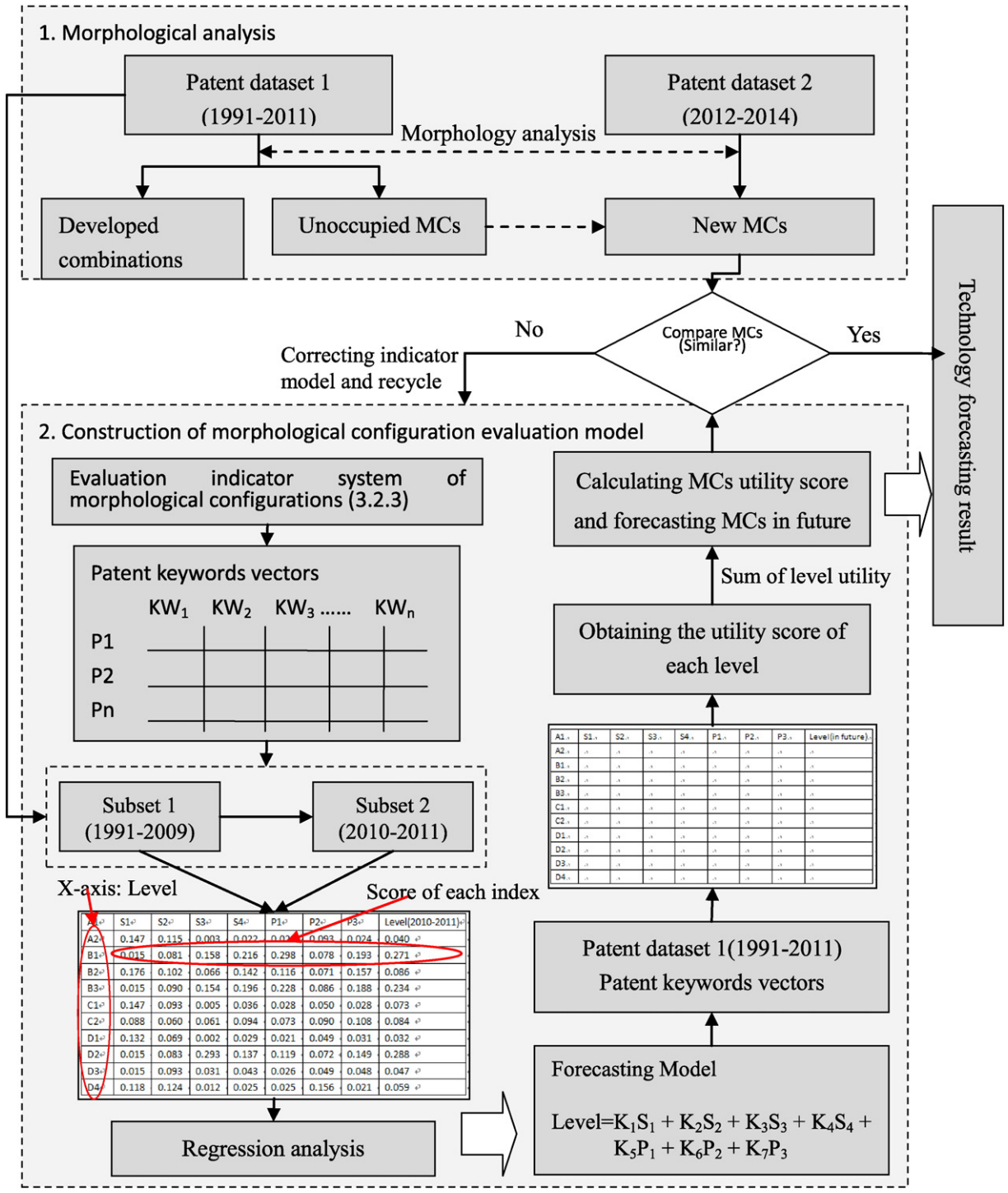


Fig. 5. Establishing and verifying morphological configuration evaluation model for technology forecasting.

patent-keywords matrix. Based on patent-keywords matrix and network, we removed the keywords that do not co-occur and then selected the patents that include two or more keywords in the abstract to obtain 199 patents. Part of the patent-keywords network is shown in Fig. 6.

The example to screen patent is as follows: from Fig. 6, three keywords co-occur in patent US7351359, including "sulfonic acid", "organic solvent" and "titanium dioxide". Therefore, the patent US7351359 is what we want. Oppositely, there is only one keyword "metal complex" in patent US627856, which has no co-occur. Therefore, the patent should be left out.

4.2. Establishing morphological structure based on SAO analysis

Using the Stanford Parser, we extracted SAO structures from the abstracts of 199 patents. As mentioned previously, the morphological structure used in this research is a hierarchy technology tree. Therefore, we are primarily interested in SAO structures that have hierarchical

Table 3
Technology keywords (30 of 166 keywords).

Keywords	Frequency	Keywords	Frequency	Keywords	Frequency
Electrolyte	1272	Organic solvent	224	Glass substrate	144
Counter electrode	842	Anode	177	Ionic liquid	136
Semiconductor	368	Zinc oxide	177	Conductive film	128
Sensitizing dye	352	Titanium oxide	176	Semiconductor fine particles	119
Electrolyte solution	336	Organic thin-film solar cell	167	Lithium ion battery	118
Electrodes	313	Porous semiconductor layer	163	Electrolyte layer	117
Semiconductor electrode	300	Pigment sensitizing type solar cell	162	Second electrode	116
Titanium dioxide	267	Organic solar cell	159	First electrode	111
Conductive substrate	235	Electrolyte liquid	151	Sealing material	109
Metal oxide	233	Porous semiconductor	150	Photosensitizer	108

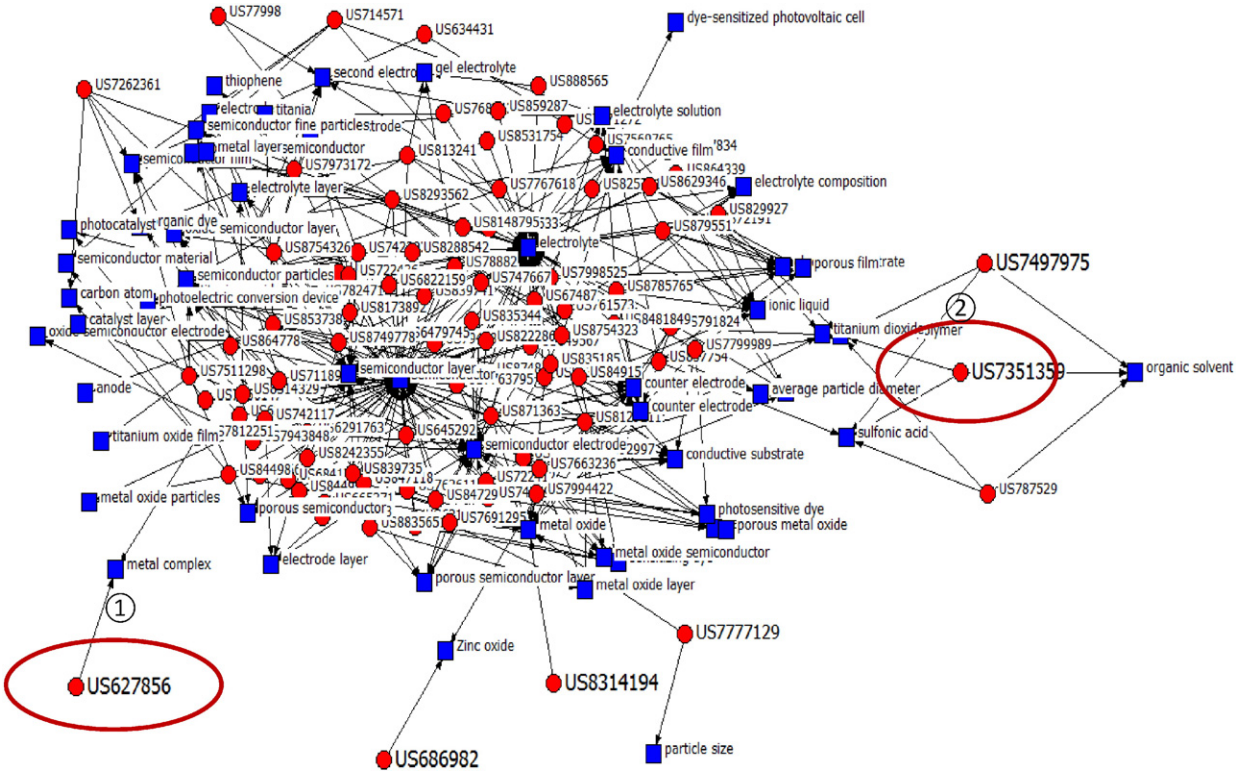


Fig. 6. Patent-keywords network (100 of 1879 patent).

relationships. The basis of judgment is the partitive verbs (Table 4) extracted from the action components of the SAO structures. We obtained 155 SAO structures. Then, SAO structures were connected automatically with reference to the corresponding words from the S/O components to construct an SAO chain (Table 5).

According to the hierarchy rules shown in Table 1, we can identify the hierarchical relationship of keywords. Several SAO chain extracted from abstracts of patent is shown in Table 6, and the hierarchical relationship of keywords based on SAO chain listed in Table 6 is illustrated in Fig. 7.

Table 4
Partial words for hierarchy relationship judgment.

Partative type verb	Representing containment relationship between products or technologies	Contain, form, include, consist, has/have, adsorbed on/with, coat with, support, comprise, compose, covering, be formed, be manufactured, be obtained, be produced, be selected, be included, be made of, be composed of

Similarly, we establish a technology morphological structure based on the 155 SAO structures. After artificially merging and filtering keywords, we finally obtain technology morphology for DSSC technology (Table 7).

4.3. Morphology analysis and technology morphology forecasting model based on historical data

According to the technology morphological structure described in Section 4.2, DSSC technology has 81 ($3 \times 3 \times 3 \times 3 = 81$) distinct combinations. We transform patent documents into keywords vectors and obtain the DSSC technology morphology vectors (Table 8). In this analysis, 253 patents with a complete morphological structure and 14 morphological configurations are identified prior to 2011 (Table 9). The frequency and time series distribution of the 14 morphological configurations are shown in Fig. 8. As can be seen, for DSSC technology, 67 (81–14 = 67) unoccupied combinations are evident.

After identifying the developed and unoccupied configurations, the next step is to evaluate the morphological configurations for technology forecasting. Before calculating the utility of the morphological configurations, each level of the morphological structure should be

Table 5
Part of SAO chains and hierarchical relationship of S/O.

Patent no	Type	S	A	O	Level 1	Level 2	Level 3
JP2007265694-A	3.1	Photoelectrode	Comprising	Surface modification compound	Photoelectrode	Surface modification compound	Metal oxide
JP2007265694-A		Surface modification compound	Supported on	Porous membrane	Porous membrane		
JP2007265694-A		Porous membrane	Containing	Metal oxide and ruthenium dye			Ruthenium dye
JP2007265694-A		Metal oxide	Such as	Titania			
JP2009105042-A	3.1	A translucent conductive paint	Contains	Electroconductive oxide powder	A translucent conductive paint	Electroconductive oxide powder	Indium oxide doped with metal oxide
JP2009105042-A		Electroconductive oxide powder	Comprising	Indium oxide doped with metal oxide			
JP2009059796-A	3.1	A photoelectric transducer	Consists of	A mixed conductor film	A photoelectric transducer	A mixed conductor film	An organic pigment
JP2009059796-A		A mixed conductor film	Containing	An organic pigment			
JP2009059796-A		Organic pigment	Having	Porphyrin			Porphyrin
JP2008287893-A	1.2	A catalyst electrode	Containing	Platinum	Porous titanium oxide layer	A catalyst electrode	Platinum
JP2008287893-A		A catalyst electrode	Is formed in	The surface of porous titanium oxide layer			
JP2008198482-A	3.1	A dye-sensitized solar cell	Includes	Electrolyte layer	A dye-sensitized solar cell	Electrolyte layer	Ionic liquid
JP2008198482-A		Electrolyte layer	Containing	Ionic liquid			
JP2008198482-A		Ionic liquid	As	Solvent			Solvent
TW201001724-A	1.1	A dye-sensitized solar cell	Having	Anode	A dye-sensitized solar cell	Anode	A double-layer nanotube structure
TW201001724-A		A double-layer nanotube structure	As	An anode for the cell			
TW201001724-A	1.1	A nano mold	Formed by	Growing aluminum oxide nanotubes	A nano mold	Growing aluminum oxide nanotubes	Titanium dioxide nanotubes
TW201001724-A		Titanium dioxide nanotubes	Are formed into	The aluminum oxide nanotubes			
BR200910292-A2	3.3	A dye-sensitized solar cells	Comprises	A gel polymer electrolyte, a counter electrode, a photo-film electrode	A dye-sensitized solar cells	A gel polymer electrolyte	
BR200910292-A2		A counter electrode	Coated with	Platinum or carbon nanotubes		A counter electrode	platinum or carbon nanotubes
BR200910292-A2		A photo-film electrode	Composed of	Titanium oxide		A photo-film electrode	titanium oxide
JP10112337-A	3.3	The semiconductor electrode	Has	Pigments	Semiconductor electrode	Pigments	
JP10112337-A		Pigments	Are included in	A metal oxide film	A metal oxide film		
JP2007299671-A	3.1	Transparent electrode	Containing	Indium tin oxide	The photoelectrode	Transparent electrode	indium tin oxide
JP2007299671-A		The photoelectrode	Has	Transparent electrode			
JP2007128757-A	1.1	The dye sensitizing solar cell	Has	A transparent semiconductor electrode	The dye sensitizing solar cell	A transparent semiconductor electrode	
				A pigment		A pigment	
				Gel-form or solid state electrolyte		Gel-form or solid state electrolyte	
				A counter electrode		A counter electrode	

evaluated. Here, we use regression analysis to build a technology morphology forecasting model. We use historical data (1991–2011) to build the forecasting model and patents issued between 2012 and 2014 to evaluate the model.

Based on the evaluation index system presented in Section 3.2.3, we identify the relationships between levels and indexes. First, we calculate the value of each evaluation index using patent data from 1991 to 2009, which is used as the independent variable. The occurrence frequency of

Table 6
Four example SAO chains.

Patent no	Type	S	A	O
BR200910292-A2	3.3	A dye-sensitized solar cell	Comprises	A gel polymer electrolyte, a counter electrode, a photo-film electrode
BR200910292-A2		A counter electrode	Coated with	Platinum or carbon nanotubes
BR200910292-A2		A photo-film electrode	Composed of	Titanium oxide
JP10112337-A	3.3	The semiconductor electrode	Has	Pigments
JP10112337-A		Pigments	Are included in	A metal oxide film
JP2007299671-A	3.1	Transparent electrode	Containing	Indium tin oxide
JP2007299671-A		The photoelectrode	Has	Transparent electrode
JP2007128757-A	1.1	The dye sensitizing solar cell	Has	Electrode
				A pigment
				Gel-form or solid state electrolyte
				A counter electrode

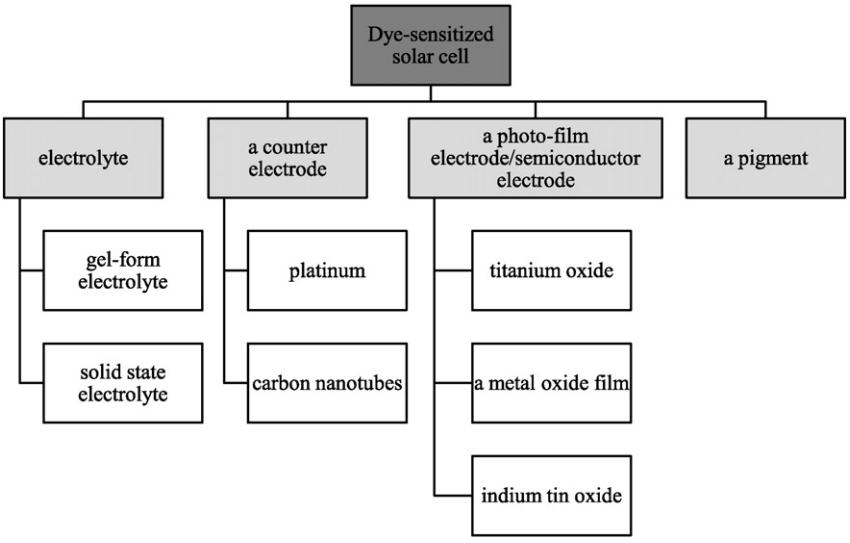


Fig. 7. Hierarchy structure obtained from SAO chain structures in Table 6.

Table 7
Technology morphological structure based on SAO chains.

Product	Components (attributives)	Materials (levels)	Keywords
Dye-sensitized solar cells	A(semiconductor oxide film)	A1(TiO_2 film)	Titanium oxide, TiO_2, titanium dioxide
		A2(laminated film)	Laminated film, polyester film
		A3(ZnO doped film)	Zinc oxide, ZnO
	B(sensitizing dye)	B1(inorganic dye)	Ruthenium, metalloporphyrin, quantum dot, phthalocyanine, porphyrin dye
		B2(natural dye)	Pigment, coumarin dye, natural dye
		B3(organic synthetic dye)	Organic dye, synthetic dye
	C(electrolyte)	C1(liquid electrolyte)	Liquid electrolyte, electrolyte solution
		C2(gel electrolyte)	Gel electrolyte, quasi-solid electrolyte, gel polymer, gelled polymer electrolyte, gel-form
		C3(solid electrolyte)	Solid polymer, solid state electrolyte, solid electrolyte
	D(counter electrode)	D1(carbon electrode)	Graphite fiber, graphene, carbon electrode
		D2(platinum electrode)	Platinum electrode, platinum
		D3(polymer electrode)	Polymer electrode, polythiophene electrode, polyaniline electrode, polypyrrole electrode

Table 8
Keyword frequency vectors of technology morphological structure (20 of 1879 patents).

Appl.year	Pub.Year	Patent no	Semiconductor oxide film			Sensitizing dye			Electrolyte			Counter electrode		
			TiO_2 film	Laminated Film	ZnO	Inorganic dye	Natural dye	Organic synthetic dye	Liquid electrolyte	Gel electrolyte	Solid electrolyte	Carbon electrode	Platinum electrode	Conducting polymer
1990	1991	US5017449	2	0	0	1	0	0	8	0	0	0	2	0
1994	1996	US5506420	1	0	1	1	0	0	0	0	2	0	1	0
1995	1998	US5737114	3	0	0	3	0	0	2	1	0	0	0	2
1998	2000	US6075203	29	0	0	1	0	0	4	0	0	0	1	0
1998	2000	US6084176	4	0	0	6	0	0	8	1	4	0	2	0
1999	2000	US6127127	2	0	1	1	0	0	1	0	0	2	0	0
1999	2001	US6239355	3	0	0	1	0	0	2	0	2	0	1	0
2000	2001	US6274806	3	0	1	1	0	0	5	0	0	0	33	0
1999	2001	US6278056	5	0	1	4	0	3	13	0	1	0	3	0
2000	2001	US6281429	3	0	0	3	0	0	8	1	1	0	2	0
2000	2001	US6291763	7	0	2	4	0	0	12	0	0	0	4	0
2000	2001	US6300559	101	0	0	9	0	0	1	0	0	0	7	0
1999	2002	US6340544	1	0	2	1	0	0	10	0	0	0	2	0
2000	2002	US6350946	8	0	2	4	0	0	12	0	0	0	2	0
2000	2002	US6387625	2	0	1	1	0	0	1	0	0	2	0	0
2000	2002	US6407330	2	0	0	7	0	0	1	0	0	0	1	0
2000	2002	US6420648	2	0	0	7	0	0	1	0	0	0	1	0
2000	2002	US6444189	30	0	0	1	0	0	4	0	0	0	1	0
2000	2002	US6469243	18	0	0	6	0	0	10	0	0	0	10	0
2001	2002	US6479745	26	0	0	6	0	0	33	0	1	0	5	0

Table 9
Developed morphological configurations prior to 2011.

Developed morphological configurations prior to 2011	Frequency	Patent no
Laminated film-inorganic dye-liquid electrolyte-platinum electrode	1	US8039735
TiO_2 film-inorganic dye-liquid electrolyte-carbon electrode	4	US6127127 US6387625 US7279570 US7291727
TiO_2 film-inorganic dye-liquid electrolyte-conducting polymer	1	US5737114
TiO_2 film-inorganic dye-liquid electrolyte-platinum electrode	82	US5017449 US6075203 US6084176 US6239355 US6274806
TiO_2 film-inorganic dye-solid electrolyte-platinum electrode	5	US5506420 US6706963 US6991958 US7880082 US8003884
TiO_2 film-natural dye-liquid electrolyte-platinum electrode	5	US7202412 US7224036 US7422923 US7825330 US7932465
TiO_2 film-organic synthetic dye-liquid electrolyte-carbon electrode	2	US7420117 US7973172
TiO_2 film-organic synthetic dye-liquid electrolyte-platinum electrode	23	US6278056 US6639073 US6683361 US6822159 US7078613
TiO_2 film-organic synthetic dye-solid electrolyte-platinum electrode	1	US6929970
ZnO-inorganic dye-gel electrolyte-platinum electrode	1	US7927913
ZnO-inorganic dye-liquid electrolyte-platinum electrode	7	US6340544 US7481914 US7671272 US7709824 US7812251
ZnO-inorganic dye-solid electrolyte-platinum electrode	1	US7968792
ZnO-natural dye-liquid electrolyte-platinum electrode	1	US6596078
ZnO-organic synthetic dye-liquid electrolyte-platinum electrode	4	US6936761 US7468146 US7884218 US7910824

each level in existing morphological configurations from 2010 to 2011 is the dependent variable. After consistent processing and normalizing the value of the indexes, we obtain normalized data (Table 10). The index value of each component from 1991 to 2009 is shown in Fig. 9. We then use SPSS software to perform regression analysis. The results of regression analysis are in Table 11.

The quantitative results indicate that the regression analysis model reached a correlation coefficient (R) of 0.982 and determination coefficient (R^2) of 0.964. These results validate the proposed forecasting model. The probability value for the F statistic is 0.024. Therefore, we conclude that Y and the independent variables (TI1, TI2, TI3, TI4, TI5, TI6, TP1 and TP2) correlate at significance levels less than 0.05. Thus, based on the coefficients table, the utility model we built is as follows:

$$Y = -0.006TI_1 + 0.006TI_2 + 0.986TI3 - 0.573TI_4 + 0.923TI_5 - 0.094TI_6 - 0.613TP_1 + 0.27TP_2. \quad (1)$$

4.4. Forecasting future technology morphological configurations

To forecast the future utility level, we first calculate independent variables using patent data from 1991 to 2011 to obtain the value of each index (Table 12).The distribution of index values for each component from 1991 to 2011 is shown in Fig. 10.

We then use the forecasting model discussed in Section 4.3 to calculate the value of Y, which is the future utility of levels. The results are shown in Table 13. Thus, the utility of morphological configurations can be obtained by combining the utility of levels. The top 20 morphological configurations are shown in Table 14.

From Table 13, the components with the highest utility are liquid electrolyte, TiO_2 film, platinum electrode and inorganic dye. Simultaneously, from Table 14, the configuration with the highest utility is the TiO_2 film-inorganic dye–liquid electrolyte–platinum electrode, which was developed in 71 patents and was public first in 1991 (Table 8). This indicates that these four technologies and their combination are the basic and important technologies for DSSC. The next highest utility configurations are the TiO_2 film-organic synthetic dye–liquid electrolyte–platinum electrode, the ZnO-inorganic dye–liquid electrolyte–platinum electrode and the TiO_2 film-inorganic dye–liquid electrolyte–carbon electrode. These morphological configurations can be considered DSSC technology hotspots, in which there is significant opportunity and intense competition for a middle or short term.

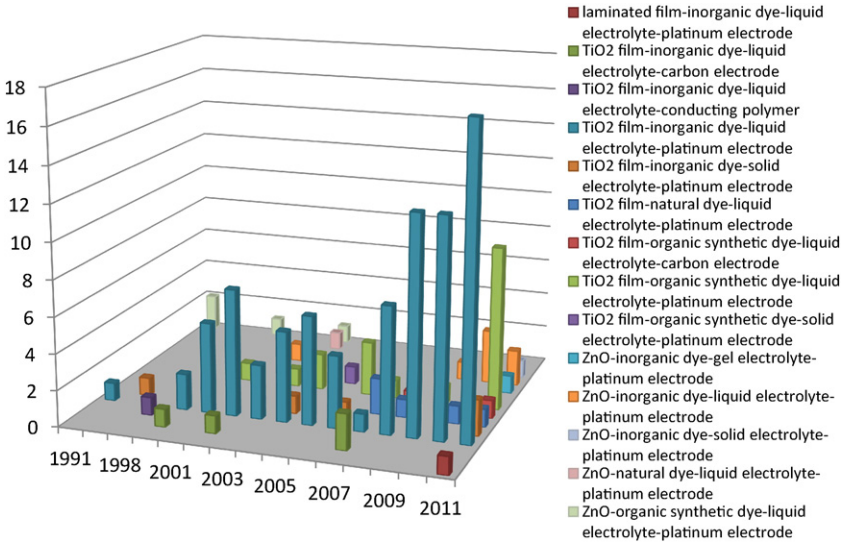


Fig. 8. Distribution of developed morphological configurations (1991–2011).

Table 10
Normalized data for regression analysis.

Level	TI1	TI2	TI3	TI4	TI5	TI6	TP1	TP2	Y
Laminated film	0.147	0.115	0.004	0.020	0.025	0.036	0.021	0.037	0.004
TiO_2 film	0.015	0.081	0.166	0.260	0.326	0.023	0.240	0.223	0.207
ZnO	0.176	0.102	0.048	0.116	0.110	0.024	0.132	0.152	0.039
Inorganic dye	0.015	0.090	0.163	0.220	0.230	0.024	0.201	0.213	0.168
Natural dye	0.147	0.093	0.004	0.050	0.031	0.013	0.032	0.026	0.009
Organic synthetic dye	0.088	0.060	0.054	0.060	0.060	0.030	0.075	0.094	0.073
Gel electrolyte	0.132	0.069	0.002	0.034	0.023	0.016	0.042	0.022	0.004
Liquid electrolyte	0.015	0.083	0.296	0.137	0.118	0.022	0.146	0.132	0.233
Solid electrolyte	0.015	0.093	0.035	0.049	0.027	0.014	0.053	0.032	0.013
Carbon electrode	0.118	0.124	0.010	0.004	0.014	0.607	0.002	0.020	0.004
Conducting polymer	0.118	0.007	0.002	0.001	0.000	0.053	0.001	0.000	0.000
Platinum electrode	0.015	0.084	0.215	0.049	0.037	0.136	0.055	0.047	0.246

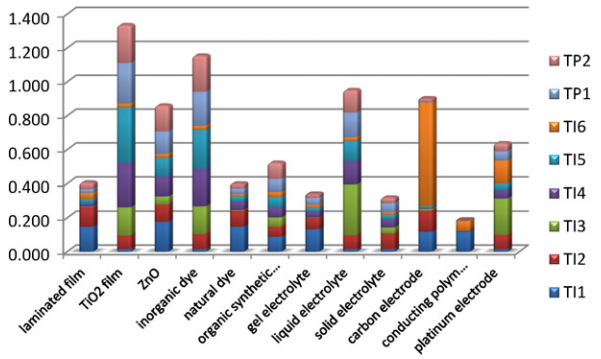


Fig. 9. Index values of each component (1991–2009).

As can be seen in Table 14, the most potent and unoccupied configuration is the TiO_2 film-inorganic dye–gel electrolyte–platinum electrode, which is the most promising future configuration. The next highest unoccupied configuration is the TiO_2 film-organic synthetic dye–liquid electrolyte-conducting polymer, followed by the TiO_2 film-natural dye–liquid electrolyte–carbon electrode, the laminated film-organic synthetic dye–liquid electrolyte–platinum electrode and the ZnO-inorganic dye–liquid electrolyte–carbon electrode. These morphological configurations represent emerging technologies that require ongoing development in future.

Besides, by combining the result of Table 13, future technology innovation and new product development should focus on ZnO doped film, organic synthetic dye, solid electrolyte and carbon electrode, which are the second highest utility components.

We also compared the results of the proposed method with actual data from 2012 to 2014 (Table 15). We found that the utility of all morphological configurations from 2012 to 2014 are in the top 32. Moreover, the top two utility score morphological configurations appeared frequently in 2012–2014. Note that the most promising forecast combination also appears in 2012–2014 and the distribution of morphological configurations we predict are also close to the fact (Fig. 11). Accordingly, the validity of the proposed method has been demonstrated.

5. Conclusion and future work

We have proposed an improved morphology analysis method based on SAO semantic analysis and bibliometric methods for technology forecasting and technology innovation.

To address the limitations of keyword-based approaches in previous research, we have combined SAO semantic analysis and keywords frequency vectors analysis to construct morphological structure, which is a key improvement in the proposed method. SAO semantic analysis can clearly represent how keywords are related and can identify the purpose of a given technology. Thus, when compared with previous research, SAO-based morphology analysis method considers the relationships between keywords, which can compensate for the limitations of keyword-based approaches and build morphological structure more objectively.

To evaluate and forecast future morphological configurations, we have combined patent analysis and bibliometrics analysis to build a systematic and comprehensive technology evaluation index system. Using regression analysis, a morphology forecasting model is constructed based on historical data. Finally, to validate the proposed method, an empirical study was conducted using DSSC technology as an example. We also used the latest patent data 2012 to 2014 to verify the prediction results.

The contributions of the proposed method are summarized as follows. First, the proposed method makes full use of objective data, which reduces the dependence on expert, time and cost. In the construction of the morphological structure, we use SAO semantic analysis to identify the relationships of keywords rather than expert knowledge. Rather than solely relying on expert scoring, we employ a systematic evaluation index system to evaluate morphological configurations. The system has sub-indexes that can be used to evaluate the importance and potential of sub-technologies. Note that the proposed method can be used for different technologies, especially emerging technology.

Second, the proposed method improves morphology analysis method and can effectively forecast future morphological configurations and the direction of technology change. In the method, we combined patent analysis and bibliometrics analysis to build a systematic and comprehensive technology evaluation index system. In Section 4.4, the correlation coefficient (R) indicator and determination coefficient indicator of

Table 11
Regression analysis results.

Model	R	R square	Adjusted R square	Std. error of the estimate	Change statistics					Durbin–Watson
					R square change	F change	df1	df2	Sig. F change	
1	0.982	0.964	0.866	0.036	0.964	9.918	8	3	0.043	1.622

a. Predictors: (constant), TP2, TI6, TI2, TI1, TI3, TI5, TP1, TI4.
b. Dependent variable: Y.

Table 12
Normalized data to calculate utility of each level.

Level	TI1	TI2	TI3	TI4	TI5	TI6	TP1	TP2
Laminated film	0.147	0.115	0.003	0.022	0.024	0.093	0.024	0.037
TiO_2 film	0.015	0.081	0.158	0.216	0.298	0.078	0.193	0.165
ZnO	0.176	0.102	0.066	0.142	0.116	0.071	0.157	0.133
Inorganic dye	0.015	0.090	0.154	0.196	0.228	0.086	0.188	0.208
Natural dye	0.147	0.093	0.005	0.036	0.028	0.050	0.028	0.023
Organic synthetic dye	0.088	0.060	0.061	0.094	0.073	0.090	0.108	0.131
Gel electrolyte	0.132	0.069	0.002	0.029	0.021	0.049	0.031	0.016
Liquid electrolyte	0.015	0.083	0.293	0.137	0.119	0.072	0.149	0.148
Solid electrolyte	0.015	0.093	0.031	0.043	0.026	0.049	0.048	0.029
Carbon electrode	0.118	0.124	0.012	0.025	0.025	0.156	0.021	0.051
Conducting polymer	0.118	0.007	0.001	0.001	0.001	0.136	0.001	0.001
Platinum electrode	0.015	0.084	0.214	0.059	0.042	0.071	0.053	0.058

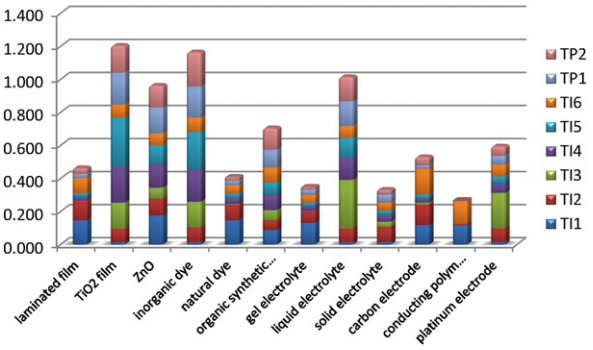


Fig. 10. Distribution of index values for each component (1991–2011).

our forecasting model can be accepted. Besides, by checking the results we predict against actual data from 2012 to 2014, we found that future promising morphological configurations and the direction of technology development we predict are close to the fact. Therefore, the proposed method is an improved morphology analysis method and can effectively forecast future morphological configurations and the direction of technology development.

Third, the proposed method can be used to dynamically monitor and forecast technology development. In the proposed method, the key to technology forecasting is construction of morphological structure and the model of morphological configurations evaluation. As technology continues to develop, we simply need to add new keywords and relationships relative to specific technology and update the value of each level. Thus, a new technology forecasting model can be constructed. The proposed method can then be used to identify current developed morphological configurations and forecast future technology morphological configurations further.

However, some limitations and future developments must be considered. First, the proposed method derives the SAO structures from abstracts. Although abstracts can summarize the main content of patents and are used by many researchers to perform SAO analysis, they may omit certain technological details. To improve results, both the abstract and the full text should be considered when extracting SAO structures. Second, although we can effectively forecast future morphological configurations based on the proposed evaluation index system, the evaluation index could be improved further. Future research can take technical performance and technical risk into consideration. Finally, the results presented in this study may be limited because the empirical study focused on a single technology field. In future, we plan to apply the proposed method to other fields and test its effectiveness further.

Table 13
Utility levels for forecasting future technology morphological configurations (after 2011).

Attribute	Level	Utility	Attribute	Level	Utility
A(semiconductor oxide film)	A1(laminated film)	−0.011	C(electrolyte)	C1(gel electrolyte)	−0.0224
	A2(TiO_2 film)	0.2262		C2(liquid electrolyte)	0.2619
	A3(ZnO doped film)	0.0122		C3(solid electrolyte)	0.0030
B(sensitizing dye)	B1(inorganic dye)	0.1824	D(counter electrode)	D1(carbon electrode)	−0.0001
	B2(natural dye)	−0.0142		D2(conducting polymer)	−0.0196
	B3(organic synthetic dye)	0.0289		D3(platinum electrode)	0.1920

Table 14
Top 20 future morphological configurations.

Rank	Morphological configurations	Utility	Status
1	TiO_2 film-inorganic dye-liquid electrolyte-platinum electrode	0.862	Developed
2	TiO_2 film-organic synthetic dye-liquid electrolyte-platinum electrode	0.709	Developed
3	TiO_2 film-inorganic dye-liquid electrolyte-carbon electrode	0.670	Developed
4	TiO_2 film-natural dye-liquid electrolyte-platinum electrode	0.666	Developed
5	TiO_2 film-inorganic dye-liquid electrolyte-conducting polymer	0.651	Developed
6	ZnO-inorganic dye-liquid electrolyte-platinum electrode	0.648	Developed
7	Laminated film-inorganic dye-liquid electrolyte-platinum electrode	0.626	Developed
8	TiO_2 film-inorganic dye-solid electrolyte-platinum electrode	0.604	Developed
9	TiO_2 film-inorganic dye-gel electrolyte-platinum electrode	0.578	Unoccupied
10	TiO_2 film-organic synthetic dye-liquid electrolyte-carbon electrode	0.517	Developed
11	TiO_2 film-organic synthetic dye-liquid electrolyte-conducting polymer	0.497	Unoccupied
12	ZnO-organic synthetic dye-liquid electrolyte-platinum electrode	0.495	Developed
13	TiO_2 film-natural dye-liquid electrolyte-carbon electrode	0.474	Unoccupied
14	Laminated film-organic synthetic dye-liquid electrolyte-platinum electrode	0.473	Unoccupied
15	ZnO-inorganic dye-liquid electrolyte-carbon electrode	0.456	Unoccupied
16	TiO_2 film-natural dye-liquid electrolyte-conducting polymer	0.454	Unoccupied
17	ZnO-natural dye-liquid electrolyte-platinum electrode	0.452	Developed
18	TiO_2 film-organic synthetic dye-solid electrolyte-platinum electrode	0.450	Developed
19	ZnO-inorganic dye-liquid electrolyte-conducting polymer	0.437	Unoccupied
20	Laminated film-inorganic dye-liquid electrolyte-carbon electrode	0.434	Unoccupied

Table 15
Morphological configurations developed in 2012–2014.

Forecast rank	Morphological configurations in 2012–2014	Frequency	Real rank
1	TiO_2 film-inorganic dye-liquid electrolyte-platinum electrode	38	1
2	TiO_2 film-organic synthetic dye-liquid electrolyte-platinum electrode	31	2
6	ZnO-inorganic dye-liquid electrolyte-platinum electrode	18	3
12	ZnO-organic synthetic dye-liquid electrolyte-platinum electrode	11	4
3	TiO_2 film-inorganic dye-liquid electrolyte-carbon electrode	5	5
17	ZnO-natural dye-liquid electrolyte-platinum electrode	4	6
15	ZnO-inorganic dye-liquid electrolyte-carbon electrode	3	7
9	TiO_2 film-inorganic dye-gel electrolyte-platinum electrode	1	8
16	TiO_2 film-natural dye-liquid electrolyte-conducting polymer	1	9
4	TiO_2 film-natural dye-liquid electrolyte-platinum electrode	1	10
10	TiO_2 film-organic synthetic dye-liquid electrolyte-carbon electrode	1	11
32	ZnO-inorganic dye-gel electrolyte-platinum electrode	1	12

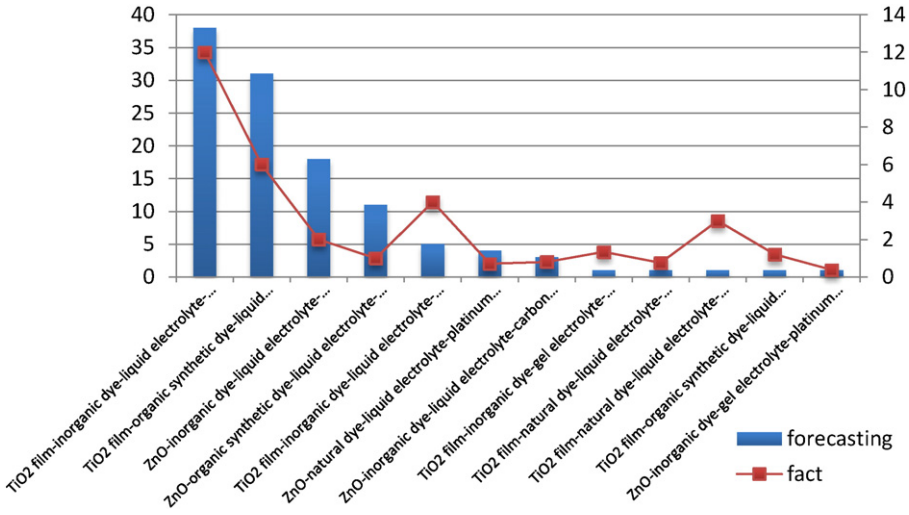


Fig. 11. Distribution of morphological configurations in 2012–2014.

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